

THE $\frac{1}{2}$ -CONJECTURE FOR q -BINOMIAL COEFFICIENTS WITH FRACTIONAL INDEX

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ABSTRACT. For a nonnegative integer k and a rational number $r \in \mathbb{Q}^+$, we define the generalized Gaussian binomial coefficient $\begin{bmatrix} r+k \\ k \end{bmatrix}_q = \frac{(q^{r+1}; q)_k}{(q; q)_k}$. When $r = a/b$ with a, b coprime positive integers and $b \geq 2$, expanding $\begin{bmatrix} r+k \\ k \end{bmatrix}_q$ via the finite q -binomial theorem produces fractional powers of q , so that $\begin{bmatrix} r+k \\ k \end{bmatrix}_q$ is a *Puiseux series* in q with nonnegative exponents; concretely it lies in $\mathbb{Q}[[q^{1/b}]]$. The notion we single out is the *integer trace* of this expansion, the subseries consisting of those terms $c_r(d) q^d$ whose exponent d is an integer, with all fractional powers discarded. This projection is not standard, and there is no a priori reason for the surviving coefficients to behave coherently as r varies. Nonetheless, ordering the family by the coefficientwise partial order leads to the $\frac{1}{2}$ -Conjecture: among all $r \in \mathbb{Q}^+$, the value $r = \frac{1}{2}$ maximizes the integer trace, in the sense that the coefficients of $\begin{bmatrix} 1/2+k \\ k \end{bmatrix}_q$ dominate those of $\begin{bmatrix} r+k \\ k \end{bmatrix}_q$ coefficientwise for every r . That so elementary a definition should single out $\frac{1}{2}$ this cleanly came as a surprise to us. We prove the conjecture in several special cases and provide further computational evidence.

1. INTRODUCTION

The *Gaussian binomial coefficients*, also called *q -binomial coefficients*, are central objects in combinatorics, number theory, and the theory of q -series (see, e.g., [1, 2, 3]). For integers $n \geq k \geq 0$ they are defined by

$$(1) \quad \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}},$$

where

$$(u; q)_n := \begin{cases} 1, & \text{if } n = 0, \\ (1-u)(1-uv) \cdots (1-uv^{n-1}), & \text{if } n \geq 1, \end{cases}$$

is the standard *q -shifted factorial* (or *q -Pochhammer symbol*) [2, p. 3]. It is classical that $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is a polynomial in q with nonnegative integer coefficients, and that it counts the number of k -dimensional subspaces of an n -dimensional vector space over $\mathbb{F}_q$ [3, Proposition 1.7.2].

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By cancellation in (1), we may write

$$(2) \quad \left[\begin{matrix} r+k \\ k \end{matrix} \right]_q = \frac{(1-q^{r+1})(1-q^{r+2}) \cdots (1-q^{r+k})}{(1-q)(1-q^2) \cdots (1-q^k)}$$

for nonnegative integers r and k . The right-hand side of (2) is meaningful for *any* $r \in \mathbb{R}$, and in particular for rational r . For example, setting $r = \frac{1}{2}$ gives

$$\left[\begin{matrix} 1/2+k \\ k \end{matrix} \right]_q = \frac{(1-q^{3/2})(1-q^{5/2}) \cdots (1-q^{1/2+k})}{(1-q)(1-q^2) \cdots (1-q^k)},$$

which is a well-defined power series in $q^{1/2}$.

In this paper, we study the extension of (1) to rational numbers. Following the notation of (2), we study the family

$$(3) \quad \left[\begin{matrix} r+k \\ k \end{matrix} \right]_q = \frac{(q^{r+1}; q)_k}{(q; q)_k}, \quad r \in \mathbb{Q}^+,$$

which coincides with the classical Gaussian binomial coefficient when r is a positive integer. For $r = a/b$ with a, b coprime positive integers and $b \geq 2$, expanding $(q^{r+1}; q)_k$ via the finite q -binomial theorem [1, Theorem 3.3] produces fractional powers of q , so $\left[\begin{matrix} r+k \\ k \end{matrix} \right]_q$ is a *Puiseux series* in q with nonnegative exponents; concretely it lies in $\mathbb{Q}[[q^{1/b}]]$. Recall that the *ring of Puiseux series* (with nonnegative exponents) over $\mathbb{Q}$ is

$$(4) \quad \mathcal{P} := \bigcup_{b \geq 1} \mathbb{Q}[[q^{1/b}]],$$

the set of formal series in q whose exponents are nonnegative rationals admitting a common denominator. As $r = a/b$ ranges over $\mathbb{Q}^+$ the denominator b is unbounded, so no single $\mathbb{Q}[[q^{1/b}]]$ contains the entire family $\left\{ \left[\begin{matrix} r+k \\ k \end{matrix} \right]_q \right\}_{r \in \mathbb{Q}^+}$; the ring $\mathcal{P}$ provides a common ambient space for all of them.

Once $\left[\begin{matrix} r+k \\ k \end{matrix} \right]_q$ is viewed inside $\mathcal{P}$, a natural question is what to do with its fractional powers. The operation we adopt is to discard them. We call the result the *integer trace* of $\left[\begin{matrix} r+k \\ k \end{matrix} \right]_q$, the subseries obtained by retaining only those terms $c_r(d) q^d$ with $d \in \mathbb{Z}_{\geq 0}$; we denote it by $\left[\begin{matrix} r+k \\ k \end{matrix} \right]_q \big|_{\mathbb{Z}}$. Equivalently, the integer trace is the image under the natural projection $\mathcal{P} \rightarrow \mathbb{Q}[[q]]$ that retains only the integer-exponent part. We write $\left[\begin{matrix} r+k \\ k \end{matrix} \right]_q \geq_q \left[\begin{matrix} s+k \\ k \end{matrix} \right]_q$ if $c_r(d) \geq c_s(d)$ for every $d \in \mathbb{Z}_{\geq 0}$, and we call $\geq_q$ the *integer trace order*; likewise $F =_q G$ means that F and G have the same integer trace. We are not aware of this projection having been studied before. Since the integer-exponent terms arise from cancellations among fractional powers, one might expect the surviving coefficients to carry little structure as r varies.

Extensive computation shows otherwise. Among all positive rational parameters r , a single value, $r = \frac{1}{2}$, dominates every other in the coefficientwise partial order on integer traces. We call this statement the $\frac{1}{2}$ -*Conjecture* (Conjecture 2.2). It is striking that so elementary an operation as deleting fractional powers should pick out $\frac{1}{2}$ in this way.

The paper is organized as follows. Section 2 introduces the $\frac{1}{2}$ -Conjecture (Conjecture 2.2) and develops the key structural observation that all integer-trace contributions from $\left[\begin{smallmatrix} 1/2+k \\ k \end{smallmatrix} \right]_q$ are nonnegative. Section 3 collects the basic properties of the integer trace order $\geq_q$ used throughout the proofs. Section 4 proves the conjecture for small values $k = 1, 2, 3$ by direct computation. Section 5 establishes the conjecture for the two infinite families $r = m + \frac{1}{2}$ (half-integers) and $r = m$ (positive integers). Finally, Section 6 presents computer-verified evidence for $r = 1/4$ and $r = 1/3$, together with stronger auxiliary conjectures on the sign of the numerator polynomials $H_k(q)$ and $U_k(q)$.

2. THE $\frac{1}{2}$ -CONJECTURE

Let k be a nonnegative integer. We begin with the finite q -binomial theorem (see, e.g., [1, Theorem 3.3]):

$$(5) \quad (u; q)_k = \sum_{s=0}^k (-1)^s \left[\begin{smallmatrix} k \\ s \end{smallmatrix} \right]_q q^{\binom{s}{2}} u^s.$$

Applying (5) with $u = q^{r+1}$ gives

$$(6) \quad (q^{r+1}; q)_k = \sum_{s=0}^k (-1)^s \left[\begin{smallmatrix} k \\ s \end{smallmatrix} \right]_q q^{E(s,r)},$$

where

$$(7) \quad E(s, r) = \binom{s}{2} + (r+1)s = \frac{s(s+1)}{2} + rs.$$

Definition 2.1. For $r \in \mathbb{Q}^+$, the *integer support* is

$$S_r := \{s \in \{0, 1, \dots, k\} \mid E(s, r) \in \mathbb{Z}\}.$$

The integer trace of $\left[\begin{smallmatrix} r+k \\ k \end{smallmatrix} \right]_q$ is determined entirely by the terms indexed by S_r , since only those contribute integer powers of q to the numerator.

For example, when $r = 1/2$, the exponent formula (7) gives

$$(8) \quad E(s, \tfrac{1}{2}) = \frac{s(s+1)}{2} + \frac{s}{2} = \frac{s^2 + 2s}{2} = \frac{s(s+2)}{2}.$$

This is an integer if and only if $s(s+2) \equiv 0 \pmod{2}$, which holds precisely when s is even. Hence

$$(9) \quad S_{1/2} = \{0, 2, 4, \dots, 2\lfloor k/2 \rfloor\}.$$

Crucially, $(-1)^s = +1$ for every $s \in S_{1/2}$, so *every contribution to the integer trace of $\left[\begin{smallmatrix} 1/2+k \\ k \end{smallmatrix} \right]_q$ is nonnegative.*

For the general case $r = a/b$ with a, b coprime positive integers and $b \geq 2$, since $s(s+1)/2$ is always an integer, the condition $E(s, a/b) \in \mathbb{Z}$ reduces to $as/b \in \mathbb{Z}$, and by $\gcd(a, b) = 1$ this is equivalent to $b \mid s$. Hence

$$(10) \quad S_{a/b} = \{0, b, 2b, \dots, b\lfloor k/b \rfloor\}.$$

The integrality condition $b \mid s$ is periodic in s with minimal period b .

Two regimes arise according to the parity of b :

- If b is odd (so $b \geq 3$), then $S_{a/b}$ contains odd values of s (e.g. $s = b$), so the signs $(-1)^s$ in (6) alternate and the integer-trace numerator has both positive and negative contributions.
- If b is even, then every $s \in S_{a/b}$ is a multiple of b and hence even, so $(-1)^s = +1$ and the integer-trace numerator has nonnegative coefficients, just as for $r = \frac{1}{2}$.

Among all $b \geq 2$, the choice $b = 2$ minimizes b and so yields the densest integer support: $S_{1/2} = \{0, 2, 4, \dots\} \cap [0, k]$ has $\lfloor k/2 \rfloor + 1$ elements, the maximum among all $S_{a/b}$. This combination of nonnegative contributions and maximal support is what motivates Conjecture 2.2.

Write the integer trace expansion as

$$(11) \quad \left[\begin{matrix} r+k \\ k \end{matrix} \right]_q \Big|_{\mathbb{Z}} = \sum_{d \geq 0} c_r(d) q^d.$$

The denominator $(q; q)_k^{-1}$ has the well-known expansion [1, Theorem 1.1]:

$$(12) \quad \frac{1}{(q; q)_k} = \sum_{n \geq 0} p_k(n) q^n,$$

where $p_k(n) \geq 0$ counts the number of partitions of n into at most k parts (equivalently, into parts each at most k). See also [3, Eq. (1.76)].

Combining (6) and (12), and selecting only integer exponents, yields the convolution formula

$$(13) \quad c_r(d) = \sum_{s \in S_r} (-1)^s \sum_{n \geq 0} \beta_{k,s}(n) p_k(d - n - E(s, r)),$$

where $\beta_{k,s}(n)$ denotes the coefficient of q^n in $\left[\begin{matrix} k \\ s \end{matrix} \right]_q$. Since $\left[\begin{matrix} k \\ s \end{matrix} \right]_q$ is a polynomial in q with nonnegative integer coefficients, we have $\beta_{k,s}(n) \geq 0$ for all n , and likewise $p_k(n) \geq 0$. The only possible source of negativity in (13) is therefore the sign $(-1)^s$.

This analysis leads to our main conjecture.

Conjecture 2.2 (The $\frac{1}{2}$ -Conjecture). *For every $r \in \mathbb{Q}^+$ and every integer $k \geq 1$,*

$$(14) \quad \left[\begin{matrix} 1/2+k \\ k \end{matrix} \right]_q \geq_q \left[\begin{matrix} r+k \\ k \end{matrix} \right]_q.$$

The conjecture asserts that $r = \frac{1}{2}$ is a *global maximizer* of the integer trace coefficients in the coefficientwise partial order on power series with nonnegative coefficients.

The following special case deserves individual attention.

Conjecture 2.3 ($\frac{1}{2}$ vs. $\frac{1}{4}$). *For all $k \geq 1$,*

$$\left[\begin{matrix} 1/2+k \\ k \end{matrix} \right]_q \geq_q \left[\begin{matrix} 1/4+k \\ k \end{matrix} \right]_q.$$

Conjecture 2.3 is a special case of Conjecture 2.2 with $r = 1/4$, but is singled out because of the proximity of $1/4$ to $1/2$ and because the computation of both sides at the level of numerator polynomials already exhibits rich combinatorial structure (see Section 6).

3. PROPERTIES OF THE INTEGER TRACE ORDER

The following properties of the integer trace order $\geq_q$ are elementary but important. We record them for use in the proofs of Sections 5 and 6.

Lemma 3.1. *The integer trace order satisfies the following properties.*

- (1) $F(q) \geq_q G(q)$ if and only if $F(q) - G(q) \geq_q 0$.
- (2) If $F(q) \geq_q G(q)$ and $c \in \mathbb{Q}^+$, then $cF(q) \geq_q cG(q)$.
- (3) If $F_1(q) \geq_q G(q)$ and $F_2(q) \geq_q G(q)$, then $F_1(q) + F_2(q) \geq_q G(q)$.
- (4) If $F_1(q) \geq_q F_2(q)$ and $F_2(q) \geq_q F_3(q)$, then $F_1(q) \geq_q F_3(q)$.
- (5) If $F(q) \geq_q G(q)$ and $k_1, k_2, \dots, k_\ell$ are positive integers, then

$$\frac{F(q)}{(1 - q^{k_1})(1 - q^{k_2}) \cdots (1 - q^{k_\ell})} \geq_q \frac{G(q)}{(1 - q^{k_1})(1 - q^{k_2}) \cdots (1 - q^{k_\ell})}.$$

- (6) If $n, m \in \mathbb{N}$ with $n \leq m$, then $\frac{q^n - q^m}{1 - q} \geq_q 0$.

Proof. Properties (1)–(4) follow directly from the definition.

(5) We first treat the case $G(q) = 0$. If $F(q) \geq_q 0$, then

$$\frac{F(q)}{1 - q^k} = F(q) \cdot \sum_{j \geq 0} q^{kj} \geq_q 0.$$

Applying this argument successively for each factor $k_1, \dots, k_\ell$ yields

$$\frac{F(q)}{(1 - q^{k_1})(1 - q^{k_2}) \cdots (1 - q^{k_\ell})} \geq_q 0.$$

The general case follows by applying this to $F(q) - G(q)$ and invoking (1).

(6) We compute

$$\frac{q^n - q^m}{1 - q} = q^n + q^{n+1} + \cdots + q^{m-1} \geq_q 0.$$

□

4. PARTIAL RESULTS: SMALL k

In this section, we prove the $\frac{1}{2}$ -conjecture when k is small.

Proposition 4.1. *The $\frac{1}{2}$ -conjecture is true for $k = 1, 2, 3$.*

Proof. For $k = 1$ and any $r \in \mathbb{Q}^+$, we compare $\left[\begin{smallmatrix} 1/2+k \\ k \end{smallmatrix} \right]_q$ and $\left[\begin{smallmatrix} r+k \\ k \end{smallmatrix} \right]_q$ directly:

$$\left[\begin{smallmatrix} 1/2+k \\ k \end{smallmatrix} \right]_q - \left[\begin{smallmatrix} r+k \\ k \end{smallmatrix} \right]_q = \frac{(1 - q^{3/2}) - (1 - q^{r+1})}{1 - q} = \frac{q^{r+1} - q^{3/2}}{1 - q} =_q \frac{q^{r+1}}{1 - q}.$$

Now

$$q^{r+1} =_q \begin{cases} 0, & \text{if } r \notin \mathbb{N}, \\ q^{r+1}, & \text{if } r \in \mathbb{N}, \end{cases}$$

so $q^{r+1} \geq_q 0$, and therefore $q^{r+1}/(1 - q) \geq_q 0$ by Lemma 3.1(5).

For $k = 2$ and $r \in \mathbb{Q}^+$,

$$\begin{aligned} \begin{bmatrix} 1/2 + k \\ k \end{bmatrix}_q - \begin{bmatrix} r + k \\ k \end{bmatrix}_q &= \frac{(1 - q^{3/2})(1 - q^{5/2}) - (1 - q^{r+1})(1 - q^{r+2})}{(1 - q)(1 - q^2)} \\ &= \frac{(q^{r+1} + q^{r+2} + q^4) - (q^{3/2} + q^{5/2} + q^{2r+3})}{(1 - q)(1 - q^2)} \\ &= {}_q \frac{q^{r+1} + q^{r+2} + q^4 - q^{2r+3}}{(1 - q)(1 - q^2)}. \end{aligned}$$

Since

$$q^{r+1} + q^{r+2} + q^4 - q^{2r+3} = {}_q \begin{cases} q^4, & \text{if } r \notin \frac{1}{2}\mathbb{N}, \\ q^4 - q^{2r+3}, & \text{if } r \in \frac{1}{2}\mathbb{N} \text{ and } r \notin \mathbb{N}, \\ (q^{r+1} - q^{2r+3}) + q^{r+2} + q^4, & \text{if } r \in \mathbb{N}, \end{cases}$$

we obtain in each case

$$\frac{q^{r+1} + q^{r+2} + q^4 - q^{2r+3}}{(1 - q)(1 - q^2)} \geq {}_q 0$$

by Lemma 3.1(6).

For $k = 3$ and $r \in \mathbb{Q}^+$,

$$\begin{aligned} \begin{bmatrix} 1/2 + k \\ k \end{bmatrix}_q - \begin{bmatrix} r + k \\ k \end{bmatrix}_q &= \frac{(1 - q^{3/2})(1 - q^{5/2})(1 - q^{7/2}) - (1 - q^{r+1})(1 - q^{r+2})(1 - q^{r+3})}{(1 - q)(1 - q^2)(1 - q^3)} \\ &= \frac{V}{(1 - q)(1 - q^2)(1 - q^3)}, \end{aligned}$$

where

$$V = {}_q q^4 + q^5 + q^6 + q^{r+1} + q^{r+2} + q^{r+3} - q^{2r+3} - q^{2r+4} - q^{2r+5} + q^{3r+6}.$$

If $r \notin \frac{1}{2}\mathbb{N}$ and $r \notin \frac{1}{3}\mathbb{N}$, then $V = {}_q q^4 + q^5 + q^6 \geq {}_q 0$.

If $r \in \frac{1}{2}\mathbb{N}$ and $r \notin \mathbb{N}$, then $V = {}_q (q^4 - q^{2r+3}) + (q^5 - q^{2r+4}) + (q^6 - q^{2r+5})$, so $V/(1 - q) \geq {}_q 0$ by Lemma 3.1(6).

If $r \in \frac{1}{3}\mathbb{N}$ and $r \notin \mathbb{N}$, then $V = {}_q q^4 + q^5 + q^6 + q^{3r+6} \geq {}_q 0$.

If $r \in \mathbb{N}$, then

$$V = {}_q q^4 + q^5 + q^6 + (q^{r+1} - q^{2r+3}) + (q^{r+2} - q^{2r+4}) + (q^{r+3} - q^{2r+5}) + q^{3r+6},$$

so $V/(1 - q) \geq {}_q 0$ by Lemma 3.1(6). $\square$

5. PARTIAL RESULTS: HALF-INTEGERS AND POSITIVE INTEGERS

Using Lemma 3.1, we prove Conjecture 2.2 in the cases where r is a half-integer $m + \frac{1}{2}$ ($m \geq 0$) or a positive integer $m \geq 1$.

Proposition 5.1. *The $\frac{1}{2}$ -conjecture holds for half-integer parameters: for every integer $m \geq 0$,*

$$(15) \quad \begin{bmatrix} 1/2 + k \\ k \end{bmatrix}_q \geq {}_q \begin{bmatrix} m + 1/2 + k \\ k \end{bmatrix}_q.$$

Proof. We prove (15) by induction on m . The base case $m = 0$ is trivial. For the inductive step it suffices to show

$$(16) \quad \left[\begin{matrix} m + 1/2 + k \\ k \end{matrix} \right]_q \geq_q \left[\begin{matrix} m + 3/2 + k \\ k \end{matrix} \right]_q.$$

Set $r = m + \frac{1}{2}$. We compute the difference $\left[\begin{matrix} r+k \\ k \end{matrix} \right]_q - \left[\begin{matrix} r+1+k \\ k \end{matrix} \right]_q$ directly:

$$\begin{aligned} & \left[\begin{matrix} r+k \\ k \end{matrix} \right]_q - \left[\begin{matrix} r+1+k \\ k \end{matrix} \right]_q \\ &= \frac{(1-q^{r+2}) \cdots (1-q^{r+k})}{(1-q)(1-q^2) \cdots (1-q^k)} [(1-q^{r+1}) - (1-q^{r+1+k})] \\ &= \frac{(1-q^{r+2}) \cdots (1-q^{r+k}) \cdot (q^{r+1+k} - q^{r+1})}{(1-q)(1-q^2) \cdots (1-q^k)} \\ (17) \quad &= \frac{(-q^{r+1})(1-q^{r+2}) \cdots (1-q^{r+k})}{(1-q)(1-q^2) \cdots (1-q^{k-1})}. \end{aligned}$$

We claim that the right-hand side of (17) is $\geq_q 0$.

Since $r = m + \frac{1}{2}$ is a half-integer, the exponents $r+1, r+2, \dots, r+k$ are half-integers as well. Expanding the product $(-q^{r+1})(1-q^{r+2}) \cdots (1-q^{r+k})$ yields terms of the form $(-1)^{j+1} q^{(j+1)r+I}$, where the leading factor $-q^{r+1}$ is mandatory and j is the number of factors $-q^{r+i}$ chosen from the remaining $k-1$ factors $(1-q^{r+i})$ (so that I is an integer). Such a term contributes to an integer power of q exactly when $(j+1)r$ is an integer, i.e. when $j+1$ is even (equivalently, j is odd). Then the sign is $(-1)^{j+1} = +1$, so its coefficient is nonnegative. Therefore

$$(-q^{r+1})(1-q^{r+2}) \cdots (1-q^{r+k}) \geq_q 0,$$

and Lemma 3.1(5), applied with denominator $(1-q)(1-q^2) \cdots (1-q^{k-1})$, yields $\left[\begin{matrix} r+k \\ k \end{matrix} \right]_q - \left[\begin{matrix} r+1+k \\ k \end{matrix} \right]_q \geq_q 0$.

By transitivity and induction, $\left[\begin{matrix} 1/2+k \\ k \end{matrix} \right]_q \geq_q \left[\begin{matrix} m+1/2+k \\ k \end{matrix} \right]_q$ for all $m \geq 0$. $\square$

Proposition 5.2. *The $\frac{1}{2}$ -conjecture holds for positive integer parameters: for every positive integer m ,*

$$(18) \quad \left[\begin{matrix} 1/2+k \\ k \end{matrix} \right]_q \geq_q \left[\begin{matrix} m+k \\ k \end{matrix} \right]_q.$$

Proof. We use the integer partition interpretations of the two sides.

Step 1: A lower bound for $\left[\begin{matrix} 1/2+k \\ k \end{matrix} \right]_q$. By definition,

$$\left[\begin{matrix} 1/2+k \\ k \end{matrix} \right]_q = \frac{(1-q^{3/2})(1-q^{5/2}) \cdots (1-q^{1/2+k})}{(1-q)(1-q^2) \cdots (1-q^k)}.$$

To extract the integer trace, we expand the numerator using (6) with $r = \frac{1}{2}$:

$$(1-q^{3/2}) \cdots (1-q^{1/2+k}) = \sum_{s=0}^k (-1)^s q^{s(s+2)/2} \left[\begin{matrix} k \\ s \end{matrix} \right]_q.$$

Only even values of s contribute integer powers, giving

$$(1 - q^{3/2}) \cdots (1 - q^{1/2+k})|_{\mathbb{Z}} = \sum_{\ell=0}^{\lfloor k/2 \rfloor} q^{2\ell(\ell+1)} \begin{bmatrix} k \\ 2\ell \end{bmatrix}_q = 1 + \sum_{\ell=1}^{\lfloor k/2 \rfloor} q^{2\ell(\ell+1)} \begin{bmatrix} k \\ 2\ell \end{bmatrix}_q \geq_q 1.$$

Therefore, by Lemma 3.1(5),

$$(19) \quad \begin{bmatrix} 1/2 + k \\ k \end{bmatrix}_q = \frac{(1 - q^{3/2}) \cdots (1 - q^{1/2+k})|_{\mathbb{Z}}}{(q; q)_k} \geq_q \frac{1}{(q; q)_k}.$$

Step 2: An upper bound for $\begin{bmatrix} m+k \\ k \end{bmatrix}_q$. By the classical combinatorial interpretation of the q -binomial coefficient [1, Theorem 1.1],

$$(20) \quad \frac{1}{(q; q)_k} = \sum_{\lambda \in P_k} q^{|\lambda|},$$

where P_k is the set of all partitions whose parts are each at most k . On the other hand,

$$(21) \quad \begin{bmatrix} m+k \\ k \end{bmatrix}_q = \sum_{\lambda \in P_{m,k}} q^{|\lambda|},$$

where $P_{m,k}$ is the set of partitions with at most m parts, each of size at most k .

Since $P_{m,k} \subseteq P_k$, we have

$$(22) \quad \frac{1}{(q; q)_k} \geq_q \begin{bmatrix} m+k \\ k \end{bmatrix}_q.$$

Combining (19) and (22) yields (18). $\square$

6. FURTHER EXAMPLES

We verify the $\frac{1}{2}$ -conjecture for $r = 1/3$ and $r = 1/4$ in the range $1 \leq k \leq 150$ with the help of a computer.

Proposition 6.1. *For every integer k with $1 \leq k \leq 150$,*

$$(23) \quad \begin{bmatrix} 1/2 + k \\ k \end{bmatrix}_q \geq_q \begin{bmatrix} 1/4 + k \\ k \end{bmatrix}_q.$$

Proof. By identity (6) with $r = 1/2$ and $r = 1/4$,

$$\begin{aligned} & \begin{bmatrix} 1/2 + k \\ k \end{bmatrix}_q - \begin{bmatrix} 1/4 + k \\ k \end{bmatrix}_q \\ &= \frac{\sum_{j=0}^k (-1)^j q^{j/2+j(j+1)/2} \begin{bmatrix} k \\ j \end{bmatrix}_q - \sum_{j=0}^k (-1)^j q^{j/4+j(j+1)/2} \begin{bmatrix} k \\ j \end{bmatrix}_q}{(1-q) \cdots (1-q^k)} \\ &= {}_q \frac{H_k(q)}{(1-q)(1-q^2) \cdots (1-q^k)}, \end{aligned}$$

where

$$(24) \quad H_k(q) = \sum_{\ell=1}^{\lfloor k/2 \rfloor} q^{2\ell(\ell+1)} \begin{bmatrix} k \\ 2\ell \end{bmatrix}_q - \sum_{\ell=1}^{\lfloor k/4 \rfloor} q^{\ell(8\ell+3)} \begin{bmatrix} k \\ 4\ell \end{bmatrix}_q.$$

We list the values of $H_k(q)$ for $k = 1, 2, 3, 4$:

$$\begin{aligned} H_1(q) &= 0; \\ H_2(q) &= q^4; \\ H_3(q) &= q^4 + q^5 + q^6; \\ H_4(q) &= q^4 + q^5 + 2q^6 + q^7 + q^8 - q^{11} + q^{12}. \end{aligned}$$

In the first three cases $H_k(q)$ has nonnegative coefficients, so $\left[\begin{smallmatrix} 1/2+k \\ k \end{smallmatrix} \right]_q - \left[\begin{smallmatrix} 1/4+k \\ k \end{smallmatrix} \right]_q \geq_q 0$ for $k = 1, 2, 3$ by Lemma 3.1(5).

For $k = 4$, the polynomial $H_4(q)$ already contains the term $-q^{11}$, so $H_k(q)$ alone is not a polynomial with nonnegative coefficients. We pair this negative term with the positive term q^8 : by Lemma 3.1(6), $(q^8 - q^{11})/(1 - q) \geq_q 0$, and the remaining terms $q^4 + q^5 + 2q^6 + q^7 + q^{12}$ have nonnegative coefficients, so $H_4(q)/[(1 - q)(1 - q^2)(1 - q^3)(1 - q^4)] \geq_q 0$.

For general $1 \leq k \leq 150$, we apply the same strategy: each negative term $-c_2 q^{d_2}$ in $H_k(q)$ is paired with a positive term $c_1 q^{d_1}$ satisfying $c_1 \geq c_2 > 0$ and $d_1 < d_2$, so that

$$\frac{c_1 q^{d_1} - c_2 q^{d_2}}{1 - q} \geq_q 0$$

by Lemma 3.1(6). Such a pairing exists for every k in this range, as confirmed by a symbolic computation program. $\square$

The corresponding code and data are available on the first author's personal webpage:

<https://irma.math.unistra.fr/~guoniu/halfconj.html>

This allows for independent verification of the computations.

Empirical evidence further suggests the following conjecture, which together with Proposition 6.1 would imply Conjecture 2.3.

Conjecture 6.2. *For all $k \geq 19$, we have $H_k(q) \geq_q 0$, where $H_k(q)$ is defined by (24).*

Proposition 6.3. *For every integer k with $1 \leq k \leq 150$,*

$$(25) \quad \left[\begin{smallmatrix} 1/2+k \\ k \end{smallmatrix} \right]_q \geq_q \left[\begin{smallmatrix} 1/3+k \\ k \end{smallmatrix} \right]_q.$$

Proof. By identity (6) with $r = 1/2$ and $r = 1/3$,

$$\begin{aligned} & \left[\begin{smallmatrix} 1/2+k \\ k \end{smallmatrix} \right]_q - \left[\begin{smallmatrix} 1/3+k \\ k \end{smallmatrix} \right]_q \\ &= \frac{\sum_{j=0}^k (-1)^j q^{j/2+j(j+1)/2} \left[\begin{smallmatrix} k \\ j \end{smallmatrix} \right]_q - \sum_{j=0}^k (-1)^j q^{j/3+j(j+1)/2} \left[\begin{smallmatrix} k \\ j \end{smallmatrix} \right]_q}{(1-q)(1-q^2) \cdots (1-q^k)} \\ &= {}_q \frac{U_k(q)}{(1-q)(1-q^2) \cdots (1-q^k)}, \end{aligned}$$

where

$$(26) \quad U_k(q) = \sum_{\ell=1}^{\lfloor k/2 \rfloor} q^{2\ell(\ell+1)} \begin{bmatrix} k \\ 2\ell \end{bmatrix}_q - \sum_{\ell=1}^{\lfloor k/3 \rfloor} (-1)^\ell q^{\ell(9\ell+5)/2} \begin{bmatrix} k \\ 3\ell \end{bmatrix}_q.$$

We list the values of $U_k(q)$ for $k = 1, 2, 3, 4$:

$$U_1(q) = 0;$$

$$U_2(q) = q^4;$$

$$U_3(q) = q^4 + q^5 + q^6 + q^7;$$

$$U_4(q) = q^4 + q^5 + 2q^6 + 2q^7 + 2q^8 + q^9 + q^{10} + q^{12}.$$

As in the proof of Proposition 6.1, the same pairing strategy (applied to the negative terms of $U_k(q)$) confirms that $\begin{bmatrix} 1/2+k \\ k \end{bmatrix}_q - \begin{bmatrix} 1/3+k \\ k \end{bmatrix}_q \geq_q 0$ for all $1 \leq k \leq 150$. The code and data are available on the same webpage. $\square$

Empirical evidence again suggests the following conjecture, which together with Proposition 6.3 would imply the $\frac{1}{2}$ -conjecture for $r = 1/3$.

Conjecture 6.4. *For all $k \geq 19$, we have $U_k(q) \geq_q 0$, where $U_k(q)$ is defined by (26).*

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